COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-DBS-007: Database Software Management

Document ID	TS-DBS-007
Title	Database Software Management
Executive Sponsor	Database Services Management, DBS Manager
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT supported agencies; employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
Version	4.0
Effective Date	07-01-2026
Last Updated	07-01-2026

1. Purpose

OIT is statutorily mandated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy the stakeholder needs surrounding Database Software Management, the following is submitted as the standard for OIT Database Services to provide the service.

The purpose of this document is to communicate the standards that the Governor's Office of Information Technology (OIT) has developed with the intent of providing enterprise standard configurations for Database Software Management. Standards are based on a best-of-value decision assessment involving multifarious factors, such as stakeholder needs, hardware configurations that are serviced and supported by OIT, as well as operational and security best practices guidelines.

The ultimate purpose of the Database Software Management Standard is to describe the minimum requirements for designing, installing, securing, monitoring, maintaining, protecting, and decommissioning any OIT supported DBMS environment.

The purpose of this policy is to document the scope of responsibilities of OIT Database Services (DBS) to ensure appropriate database software management in order to maintain and secure the database environments on DBS supported database platforms.

2. Scope

These standards apply to OIT and OIT supported agencies. This Standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Additionally, the standards assigned and prescribed by this document, instituted for use by the approval of the OIT Architecture Review Board (ARB), should be used to govern purchase and deployment decisions deemed as “go-forward” or “net-new” strategies or implementations, and are not intended for use as retroactively enforceable mandate or policy for immediate conformance (especially in cases where cost implications may be prohibitive).

Any/all standards are predicated with the assumption that there can be no point of enforcement unless and until there is an effort to re-procure, renew (contracts), reconfigure (including upgrades), or redevelop the system. At any time that a decision is made to make an investment of resources in sustaining or improving the system, conformance to existing standards should be part of the assessed level of effort for that investment activity. For guidance (or enforcement) in conformance to standards during any of these types of activities, an email request may be sent to OIT_OEA@state.co.us with the subject line “Request for Architecture Review: <projectName/systemName>”

This standard applies to OIT DBS database administrators, OIT system administrators, and database users.

For the purpose of enforcement, this standard is rated CAT 1 according to the Enforcement Category Chart.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Update effective date and/or last review date.
- Publish the ARB approved Standard to the OIT public facing website.

Architecture Review Board (ARB)

- Engage stakeholders and subject matter experts to determine standard.
- Review/Approve Standard (this document).

Database Services Management - DBS Manager

- Owner of Standard.
- Engage stakeholders and subject matter experts to determine standard.
- Update document standards and version number.
- Present to the Architecture Review Board for approval and adoption.

Additional Assignments

DBS Team

- Performs all database related installations, migrations, upgrades, patching and administration of database on the database server.
- Ensures that the data and the database structures are backed up, stored on secure storage, accessible only to authorized personnel and could be recovered when required.
- Ensures that production database environments are kept separate from non-production environments.
- Manages database user security on supported database platforms.
- Audits database activity in accordance with recommended security settings contained in the CIS database benchmark for the appropriate platform and version of database software.

SA Team

- Performs operating system or other server related software installations, upgrades and patching.
- Provides access for DBS database administrators to the “super user” (root) account on OIT supported Linux environments and Local Administrator account privileges on Windows environments.
- Manages OS user security on DBS supported database servers on behalf of the DBS.

Database User

- Requests access for only data or functionality required to perform their job duties.

All OIT staff are responsible to be aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Database Software Management

- It is the policy of OIT DBS that DBS will be responsible for and perform all Database related installations, migrations, upgrades, patches and administration of database binaries on the database server.
- It is the policy of OIT DBS that DBS is not responsible for, nor will perform operating system or other server related software upgrades or patches. However, DBS will be responsible for evaluation, coordination, and approval of any software upgrades on database servers.
- It is the policy of OIT DBS that non-database software not directly supporting database operations is not permitted on database servers.
- It is the policy of OIT DBS that new database installations, upgrades and migrations will follow the most recent Major Release supported and approved by DBS when appropriate to the client environment.
- It is the policy of OIT DBS that major releases will be maintained with the second most recent Maintenance Release Number when appropriate.
- It is the policy of OIT DBS that production database software and patches will be applied when available after appropriate evaluation in the non-production database environments by DBS.
- It is the policy of OIT DBS that all new installations, releases, and upgrades will pass User Acceptance Testing (UAT) prior to being deployed in a production environment.
- It is the policy of OIT DBS that all production database installations will reside on isolated server environments from non-production database, application, backup, redundancy and other related products unless required.
- It is the policy of OIT DBS that only Database Services approved client database software installations will be supported.

5. Compliance

All DBS team staff, OIT SA Team staff, and database users must comply with this policy in order to secure, access, store data, and maintain database servers. Non-compliance of this policy may result in corrective action.

6. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of and approved by the "Owner of Standard" listed in Section 3 (Roles and Responsibilities) of this document.

Requests may be acquired by completing the "EXCEPTION REQUEST to Technical Standard" form and including the completed form (including all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

7. References

OIT documents and policies

- CISP-003 Information Technology Security Risk Management Policy
- Enforcement Category Chart
- TS-INF-003: Identity and Access Administration

Third-party resources

- CIS Benchmarks
- CIS Oracle Database 12c Benchmark v3.0.0
- CIS Oracle Database 19c Benchmark v1.0.0
- CIS Microsoft SQL Server Benchmark

8. Definitions

Standard: A rule that specifies a particular course of action or response to a given situation. Standards are mandatory directives to carry out management's policies and are used to measure compliance with policies. Standards serve as specifications for the implementation of policies and are designed to promote implementation of high-level organization policy rather than to create new policy in themselves.

DBS: OIT Database Services Team

SA: OIT Server or System Administration Team

CIS: Center for Internet Security

CISO: Chief Information Security Officer

OEA (Office of Enterprise Architecture): The Office of Enterprise Architecture was formed as an office under OIT by Governor's Rule 8 CCR 1501-7.

OIT (Office of Information Technology): The Governor's Office of Information Technology was created and governed by C.R.S. 24-37.5, et seq.

Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification for granting the exception. Typically an exception request is subject to review and approval by an authority in the area.